

Grammar for BBF-Lang

Program $\rightarrow$ {Statement}

Statement $\rightarrow$ $\left\{ \begin{array}{l} \text{exit(Expression)} \\ \text{print(Expression)} \\ \text{Declaration} \\ \text{Assignment} \\ \text{If-Statement} \\ \text{For-Loop} \end{array} \right.$

Declaration $\rightarrow$ Identifier : ("Int" | "String") = Expression

Assignment $\rightarrow$ Identifier = Expression

If-Statement $\rightarrow$ if Condition then {Statement} {Elif-Statement} [Else-Statement] end

Elif-Statement $\rightarrow$ elif Condition then {Statement}

Else-Statement $\rightarrow$ else {Statement}

For-Loop $\rightarrow$ for Identifier in Range do {Statement} end

Range $\rightarrow$ Expression..["="]Expression

Expression $\rightarrow$ Equality

Equality $\rightarrow$ Comparison (("!=" | "==") Comparison) *

Comparison $\rightarrow$ Term ((">" | ">=" | "<" | "<=") Term) *

Term $\rightarrow$ Factor (("-" | "+") Factor) *

Factor $\rightarrow$ Unary (("/" | "*" | "%") Unary) *

Unary $\rightarrow$ ("-" | not) Unary

| Primary

Primary $\rightarrow$ IntegerLit

| StringLit

| Identifier

| "(" Expression ")"

| argv "[" Expression "]"

Condition $\rightarrow$ Expression

Identifier $\rightarrow$ Letter (Letter | Digit) *

StringLit $\rightarrow$ "" asdf

Letter $\rightarrow$ "A".."Z" | "a".."z" | "_"

Digit $\rightarrow$ "0".."9"